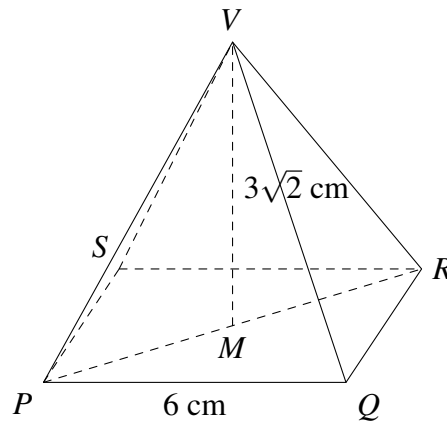


Trial

- 1** The diagram shows a pyramid with a square base $PQRS$.
 The apex V is directly above the centre M of the base.
 The sides of the square base are 6 cm and the vertical height VM is $3\sqrt{2}$ cm.



- (a) Calculate the length PR .

..... [2]

- (b) Calculate the length PM .

..... [1]

- (c) Calculate the angle between the line VP and the plane $PQRS$.

..... [2]

- 2** (a) Write as a single fraction in its simplest form.

$$\frac{3x}{4} - \frac{x-2}{3}$$

..... [2]

- (b) Fully simplify the algebraic fraction.

$$\frac{x^2 - 9}{x^2 + x - 12}$$

..... [3]

- 3 (a) Evaluate the mixed number power.

$$\left(3\frac{3}{8}\right)^{\frac{2}{3}}$$

..... [2]

- (b) Apply index laws to the expressions below.

- (i) Simplify the expression.

$$\left(\frac{64a^{12}}{b^6}\right)^{-\frac{1}{6}}$$

..... [3]

- (ii) Show that the following relationship holds for any integer m .

$$2^{m+3} + 2^m = 9 \times 2^m$$

[3]

4 Simplify the following operations completely.

(a) $2^{-4} \times 2^6$

..... [1]

(b) $\frac{6^2}{6^{-1}}$

..... [1]

(c) Show that the expression $\left(\frac{2}{3}\right)^{-2}$ reduces exactly to $2\frac{1}{4}$.

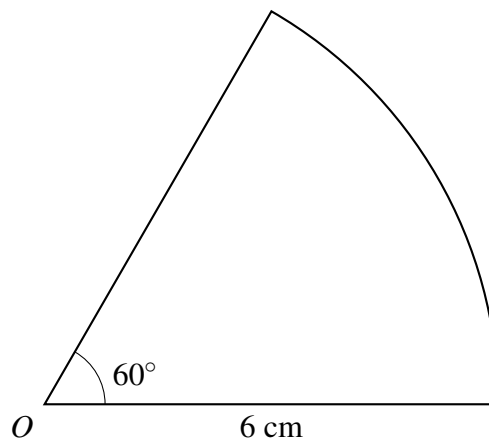
[2]

5 Make x the subject of the formula.

$$y = \frac{x+5}{x}$$

..... [3]

- 6** The diagram shows a sector of a circle with centre O and radius 6 cm.
The sector angle is 60° .



- (a)** Show that the exact arc length of the minor sector is 2π cm.

[2]

- (b)** Calculate the exact area of the major sector in terms of π .

..... [2]

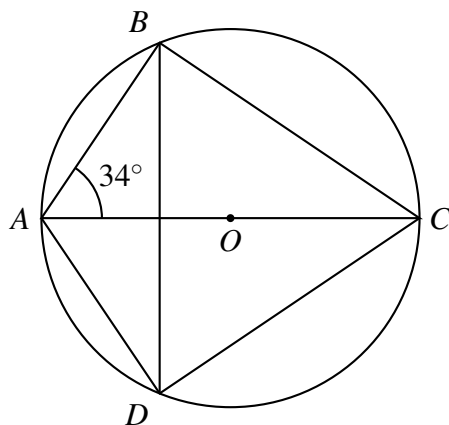
- 7 (a) A commercial transport vehicle travels the first $\frac{1}{3}$ of its total journey distance at a constant average speed of x km/h, and the remaining $\frac{2}{3}$ of the distance at a constant average speed of $2x$ km/h. Show that the average speed for the entire journey is exactly $\frac{3}{2}x$ km/h.

[3]

- (b) The total distance of the journey is 270 km. The total time taken for the journey is 3 hours and 45 minutes. Calculate the value of x .

..... [2]

- 8 Points A , B , C and D lie on a circle, centre O .
 AC is a diameter.
Line AD is parallel to line BC .
Angle $BAC = 34^\circ$.



NOT TO SCALE

- (a) Find angle ABC .

..... [1]

- (b) Find angle ACB .

..... [1]

- (c) Find angle CAD .

..... [1]

- (d) Find angle CDB .

..... [1]

9 The line L_1 has the equation $3x - 5y + 15 = 0$.

(a) Find the gradient of line L_1 .

..... [1]

(b) Line L_2 is parallel to line L_1 and passes through the point $(-10, -2)$. Find the equation of line L_2 , giving your answer in the form $y = mx + c$.

..... [2]

(c) Line L_2 crosses the y -axis at point P . Find the coordinates of P .

..... [1]

10 The line L_1 passes through the point $(2, -5)$ and is perpendicular to the line with equation

$$3x - 4y = 8$$

(a) Find the equation of the line L_1 , giving your answer in the form $ay = bx + c$, where a , b , and c are integers.

..... [3]

(b) The line L_1 crosses the x -axis at point A and the y -axis at point B . Find the coordinates of A and B .

..... [2]

(c) Calculate the exact area of the triangle OAB , where O is the origin.

..... [2]

11 Write the number 483.7062 correct to the specified degrees of accuracy.

(a) Correct to 2 decimal places.

..... [1]

(b) Correct to 3 significant figures.

..... [1]

12 (a) Expand and simplify.

$$x(x-1)(x+5)$$

..... [3]

(b) Factorise.

$$y^2 - 8y + 15$$

..... [2]

(c) Factorise completely.

$$14a^2b - 7ab^2$$

..... [2]

13 A delivery fleet vehicle was bought for 16000 AUD. The vehicle's value loses value exponentially at a rate of 50% per year.

(a) Calculate the value of the vehicle after 3 years.

..... [2]

(b) Determine the total value lost by the vehicle over these 3 years.

..... [1]

14 A continuous experimental tracking array archives variable data bins across four specific channels. The current data densities are given by:

$$A = 0.8125, \quad B = \frac{13}{16}, \quad C = 81.3\%, \quad D = \frac{41}{50}$$

- (a) Show analytically that channel A and channel B possess exact structural equivalence.

[2]

- (b) Arrange all four data density values in ascending order of magnitude. Show your supporting equivalence evaluations explicitly.

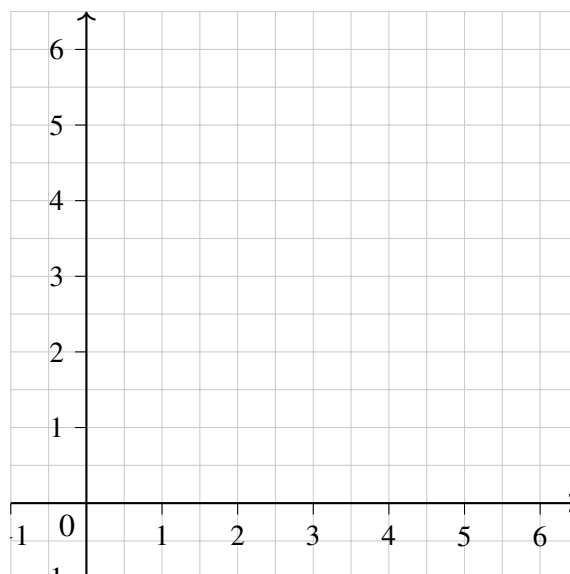
..... [3]

- 15** (a) On the grid, draw the 3 boundary lines and shade the unwanted regions to show the region defined by these 3 inequalities.

$$x \geq 2$$

$$y \geq x$$

$$y \leq 5$$



[3]

- (b) Write down all the coordinates of the points with integer values that lie entirely inside the region.

..... [2]

- 16 Triangles ABC and PQR are mathematically similar.

The length of AB is 4 cm and the length of the corresponding side PQ is 6 cm.

The area of triangle ABC is 20 cm^2 .

Calculate the area of triangle PQR .

..... [2]

- 17 The line segment joining $C(-2, -5)$ and $D(4, 3)$ forms the diameter of a circle.

- (a) Find the coordinates of the centre of this circle.

..... [2]

- (b) Show that the radius of this circle is 5.

[2]

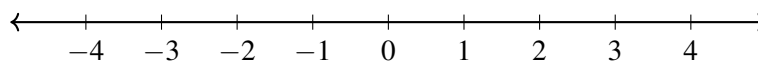
-
- 18 An inequality involving algebraic fractions is given below:

$$\frac{3(x-2)}{4} - \frac{2x+1}{3} \geq -2$$

- (a) Solve the inequality algebraically.

..... [4]

- (b) On the number line provided, represent the solution set graphically.



[2]

- (c) State the smallest integer value of x that satisfies the inequality.

..... [1]

19 A solid gold statue has a volume of 350 cm^3 .

(a) Convert the volume of the statue into cubic millimetres (mm^3).

..... [1]

(b) The density of gold is 19.3 g/cm^3 . Calculate the mass of the statue in kilograms.

..... [2]

(c) A replica of the statue is made from a polymer where every linear dimension is scaled down by a factor of $\frac{1}{5}$. Calculate the volume of the replica in cm^3 .

..... [2]

20 (a) Find the gradient and the y-intercept of the line with equation

$$3x + 4y = 12$$

(i) Gradient

..... [2]

(ii) y-intercept

..... [1]

- (b) Write down the equation of the vertical line that passes through the point $(5, -2)$.

..... [1]

- 21 (a) Arrange the following real numbers in descending order of magnitude.

$$\frac{3}{8}, \quad 0.35, \quad 32\%, \quad \frac{7}{20}$$

..... [2]

- (b) Write down the structural inequality symbol ($\geq$ or $\leq$) that correctly satisfies the given relationship.

$$\frac{12}{3} \quad \text{-----} \quad 2^2$$

..... [1]

- 22 Liam deposits 2400 EUR into a special investment account. The account accrues compound interest at a fixed rate of 5% per annum.

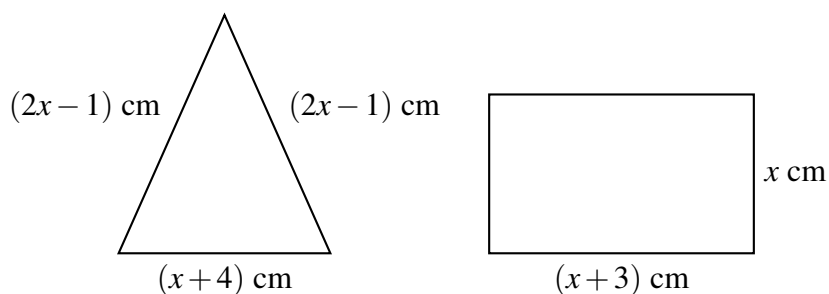
- (a) Without relying on a pre-programmed formula, show that the total value of the investment at the end of 2 years is 2646 EUR.

[2]

- (b) At the end of the third year, the total value of the investment is 2778.30 EUR. Calculate the amount of interest earned during the third year only.

..... [2]

- 23 An isosceles triangle and a rectangle are shown below.



The perimeter of the triangle is equal to the perimeter of the rectangle.

- (a) Show that the equation simplifies to find $x = 4$.

[3]

- (b) Calculate the area of the rectangle.

..... [2]

- 24 (a)** Evaluate the expression.

$$5^2 \times \sqrt[3]{8}$$

..... [2]

- (b)** Evaluate the expression.

$$10^3 - \sqrt{144}$$

..... [2]

-
- 25** An architectural draughtsman produces a layout drawing of a commercial office floor using a standard linear scale of 1 : n .

- (a)** A rectangular conference hall with real physical dimensions of 12 m by 16 m is represented on the blueprint by an area of 48 cm². Show that the value of n is exactly 200.

[3]

- (b)** Using this exact blueprint scale, calculate the map length, in centimetres, of a corridor that has an actual physical length of 54 m.

..... [2]

26 Here are the first four terms of a sequence.

7, 11, 15, 19

(a) Write down the next term of this sequence, denoted by T_5 .

..... [1]

(b) Find an expression for the n th term, T_n , of this sequence.

..... [2]

27 A group of 100 students are surveyed about their participation in Athletics (A), Badminton (B), and Cricket (C). The results show that 38 students do Athletics, 34 students do Badminton, and 41 students do Cricket. The number of students in the intersection regions are described in terms of x :

* x students participate in all three sports.

* The number of students who do Badminton and Cricket only is $2x$.

* The number of students who do Athletics and Cricket only is $x + 1$.

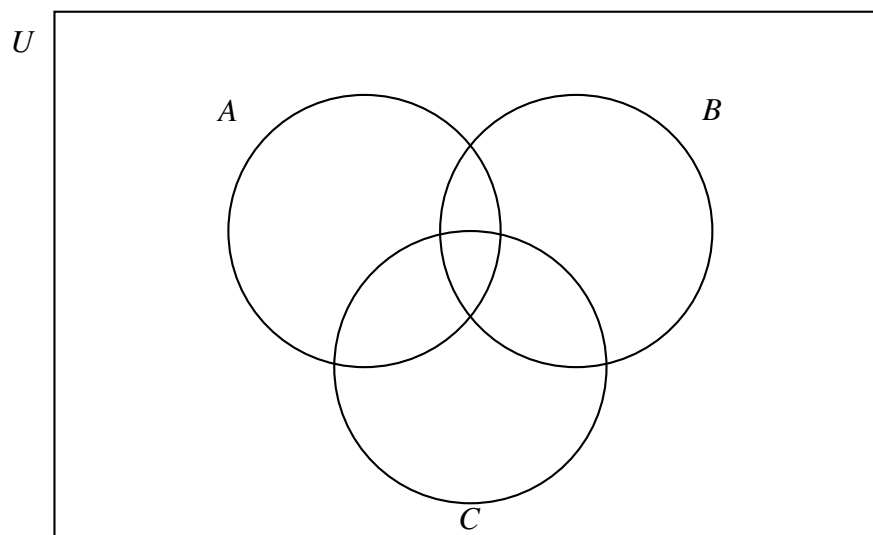
* The number of students who do Athletics and Badminton only is $2x - 3$.

* The number of students who do Cricket only is equal to the number of students who do none of the three sports.

(a) Formulate an equation in terms of x for the total number of students and show that it reduces to $155 - 11x = 100$.

[3]

(b) Complete the Venn diagram to show the exact number of students in each of the 8 regions.



[3]

- 28 (a)** Solve the quadratic equation.

$$4x^2 - 25 = 0$$

..... [2]

- (b)** Solve the equation.

$$4x^2 - 12x + 9 = 0$$

..... [2]

- 29** Calculate the exact value of the expression, giving your final answer in standard form notation.

$$\frac{(3.5 \times 10^{-5}) \times (4 \times 10^9)}{2 \times 10^2}$$

..... [3]

30 Find the gradient of the straight line passing through:

(a) the origin (0,0) and the point (3,6),

..... [1]

(b) the point (−1, −3) and the point (2,3).

..... [2]

31 (a) Simplify the expression.

$$\sqrt{75} + \sqrt{12}$$

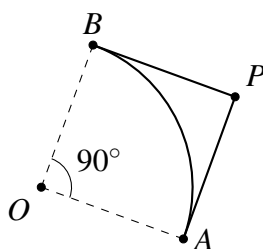
..... [2]

(b) Rationalise the denominator.

$$\frac{10}{\sqrt{5}}$$

..... [2]

- 32** A minor sector OAB of a circle, centre O , has radius 3 cm. The angle AOB is 90° . Tangents from an external point P touch the circle at A and B .



NOT TO SCALE

- (a) Find the exact area of the shape bounded by the tangents PA, PB and the minor arc AB .

..... [3]

- (b) The sector OAB forms the base of a solid which is mathematically similar to a larger solid. The corresponding radius of the base of the larger solid is 12 cm. The volume of the smaller solid is $15\pi \text{ cm}^3$. Calculate the exact volume of the larger solid.

..... [3]

- 33** (a) Write down the value of.

$$(-4) \times (-2) \times (-3)$$

..... [1]

- (b) Work out, giving your answer as a fraction in its simplest form.

$$1\frac{3}{4} + \frac{5}{8}$$

..... [2]

- 34** An aircraft leaves London (GMT) at 22 45 on a Tuesday.
The aircraft flies to Tokyo, which is in a time zone 9 hours ahead of GMT.
The total flight duration is 11 hours and 35 minutes.

- (a) Write the departure time of the aircraft using the 12-hour clock format, including p.m. or a.m.

..... [1]

- (b) Calculate the local time and day in Tokyo when the aircraft lands.

- (i) Local time

..... [2]

- (ii) Day

..... [1]

-
- 35** Two algebraic terms, P and Q , are defined below, where x , y , and z are distinct prime numbers.

$$P = 126x^4y^2z^3$$

$$Q = 540x^2y^5$$

- (a) Express 126 and 540 as products of their prime factors.

..... [2]

- (b) Find the Highest Common Factor (HCF) of P and Q in terms of x , y , and z .

..... [2]

- (c) Find the Lowest Common Multiple (LCM) of P and Q in terms of x , y , and z . Give your coefficient as an integer.

..... [2]